

Supplementary Material

Figure S1 presents the reliability of the operational ensemble for the Negative event on the held-out exploratory evaluation split. Fixed-width probability bins were used, and empty bins were omitted. The upper panel compares observed Negative frequency with mean predicted probability, whereas the lower panel reports the corresponding bin counts. Uncertainty was estimated from 2,000 stratified bootstrap resamples that preserved the reference Negative and non-Negative class counts.

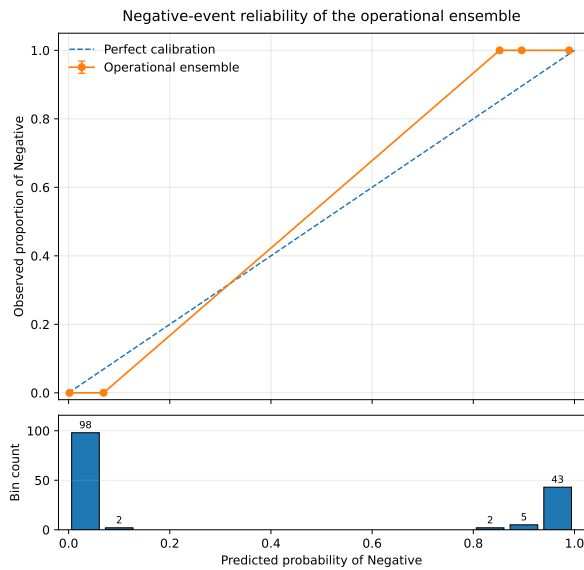


Figure S1. Negative-event reliability of the operational ensemble on the held-out exploratory evaluation split. The upper panel shows observed Negative frequency against mean predicted probability within each non-empty fixed-width bin; error bars indicate stratified-bootstrap 95% confidence intervals, and the dashed diagonal denotes perfect calibration. The lower panel reports the number of images contributing to each bin.

Table S1 summarizes the computational characteristics of the evaluated architectures and of the final operational ensemble. Reported values include trainable parameter counts, recorded training times, optimizer steps, batch-one inference latency, and peak GPU memory usage. Training times correspond to complete 20-epoch runs

averaged across the nine experiments conducted for each architecture, whereas inference measurements were obtained on an NVIDIA L4 GPU after warm-up and exclude image loading.

To complement the final ensemble results reported in the main manuscript, Table S2 summarizes the seed-specific operating policies obtained for the selected ResNet-18–Focal Loss–Vector Scaling configuration. Each row corresponds to one independently trained checkpoint, its fitted seed-specific calibrator, and the threshold frozen before evaluation on the held-out exploratory evaluation split. These results are presented to characterize stability across training seeds. The final operational ensemble described in the main manuscript instead averages the three sets of logits, fits a single Vector Scaling calibrator, and uses one frozen operating threshold.

To complement the final ensemble results reported in the main manuscript, Table S2 summarizes the seed-specific operating policies obtained for the selected ResNet-18–Focal Loss–Vector Scaling configuration. Each row reports the operating threshold, release count, coverage, observed NPV, true-negative release rate, and one-sided 95% upper confidence bound obtained for one independently trained seed. These results characterize the stability of the selected operating policy across random initializations. The final operational ensemble described in the main manuscript instead averages the three sets of logits, fits a single Vector Scaling calibrator, and uses one frozen operating threshold.

Figure S2 illustrates how observed unsafe-release risk evolves with increasing coverage during threshold selection, by showing the seed-specific risk–coverage curves obtained during threshold selection for the selected ResNet-18–Focal Loss–Vector Scaling configuration. The no-release operating point is explicitly included to illustrate the full range of feasible operating thresholds.

Table S3 reports both the conventional three-class predictions of the final ensemble before abstention and the final release/referral decisions produced after application of the selective-release rule. The operational table collapses Positive and Uncertain images into the non-Negative category

Table S1. Computational characteristics of the evaluated architectures and the final operational ensemble.

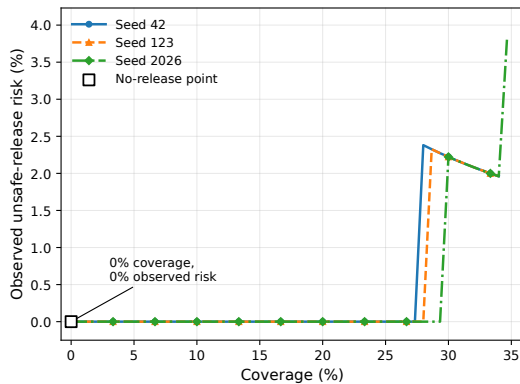
System	Trainable parameters (M)	Training time (min/run)	Optimizer steps/run	Mean latency (ms/image)	Peak inference GPU memory (MB)
ResNet-18	11.178	26.33 $\pm$ 7.53	580	2.608	62.09
EfficientNetV2-S	20.181	21.76 $\pm$ 1.85	580	18.268	94.33
Swin-T	27.522	19.87 $\pm$ 0.26	580	11.296	131.83
Three-checkpoint ResNet-18 ensemble	33.534	—	—	7.681	148.98

Training times are mean $\pm$ sample standard deviation across nine 20-epoch runs per architecture; all 27 runs required 10.19 h. Batch-one latency and peak GPU memory were measured on an NVIDIA L4 after 50 warm-up and 300 timed forward passes, excluding image loading. Training times are descriptive because runs used separate Colab sessions. The ensemble stores three ResNet-18 checkpoints and has no optimizer-step count.

Table S2. Seed-specific operating policies of the selected ResNet-18–Focal Loss–Vector Scaling configuration evaluated on the held-out exploratory evaluation split.

Seed	Threshold	Released	Coverage	Unsafe	Observed NPV	TNRR	One-sided 95% upper bound
42	0.981801	36	24.00%	0	1.0000	0.7200	0.0798
123	0.960583	41	27.33%	0	1.0000	0.8200	0.0705
2026	0.931031	45	30.00%	0	1.0000	0.9000	0.0644

Each row corresponds to one independently trained seed-specific checkpoint and its frozen seed-specific calibration policy. TNRR denotes the true-negative release rate. The upper bound is the exact one-sided 95% Clopper–Pearson upper confidence bound for the unsafe-release probability conditional on release. These rows characterize sensitivity to initialization on the same evaluation images and are not independent test estimates; the final operational ensemble uses one separate frozen threshold.

**Figure S2.** Seed-specific risk–coverage curves obtained during threshold selection for the selected ResNet-18–Focal Loss–Vector Scaling configuration. The square marker denotes the no-release operating point, corresponding to 0% coverage and 0% observed unsafe-release risk.

because neither category is eligible for automatic release.

Table S4 summarizes the number of release

Table S3. Multiclass predictions and operational release decisions of the final ensemble on the held-out exploratory evaluation split.**(a) Three-class confusion matrix before abstention**

Reference	Predicted Negative	Predicted Positive	Predicted Uncertain
Negative	50	0	0
Positive	0	45	5
Uncertain	0	6	44

(b) Release/referral table

Reference	Released	Referred
Negative	39	11
Non-Negative	0	100

Panel (a) reports three-class predictions before application of the selective-release rule. Panel (b) collapses Positive and Uncertain reference images into the operational non-Negative category and reports the final release or referral action. Only images predicted as Negative with calibrated $P(\text{Negative}) \geq \tau$ were eligible for release. The final ensemble released 39 of 150 images, referred 111 images for manual review, and produced zero observed unsafe releases.

decisions obtained for each probability condition during the original threshold-selection analysis. These counts complement the aggregate coverage results reported in the main manuscript by showing the number of automatically released images across the 27 seed-specific evaluations. Because the same images are evaluated across seeds, these values are intended as computational denominators rather than independent clinical estimates.

Table S4. Release counts by probability condition on the original threshold-selection partition.

Condition	Releases	Mean coverage	Zero-release runs
Temperature Scaling	935	23.09%	0/27
Beta Calibration	929	22.94%	0/27
Uncalibrated	929	22.94%	0/27
Vector Scaling	940	23.21%	0/27
Dirichlet Calibration	912	22.52%	0/27
Isotonic Regression	121	2.99%	24/27

Counts summarize 27 seed-specific evaluations per condition (nine architecture–loss combinations $\times$ three seeds) and are computational rather than independent clinical denominators.